A.4 Chapter 4: Refining the best result with improvement-based policies

There are two exercises in this chapter:

1. The first covers a way of refining the Probability of Improvement policy, allowing it to better explore the search space.
2. The second applies the two Bayesian optimization policies we have learned to a simulated real-world task of hyperparameter tuning.

A.4.1 Encouraging exploration with Probability of Improvement

This exercise, implemented in `CH04/02 - Exercise 1.ipynb`, walks us through how to modify Probability of Improvement to encourage exploration. There are multiple steps:

1. Recreate the Bayesian optimization (BayesOpt) loop in `CH04/01 - BayesOpt loop`, which uses the one-dimensional Forrester function as the optimization objective.
2. Before the for loop that implements BayesOpt, declare a variable named `epsilon`.

```
epsilon = 0.1
```

3. Inside the for loop, initialize the PoI policy as before, but this time specifying that the incumbent threshold, set by the `best_f` argument, is the incumbent value *plus* the value stored in `epsilon`.

```
policy = botorch.acquisition.analytic.ProbabilityOfImprovement(
    model, best_f=train_y.max() + epsilon
)
```

4. Rerun the notebook and observe whether this modification leads to better optimization performance than the original PoI policy by encouraging more exploration, as shown as Figure A.5.

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